Aditya Bidwai

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Summary: Robotics Engineer with 2+ years of experience deploying and developing software, control/motion planning algorithms for real-world robots like UGVs, UAVs, and legged robots using C++/Python, ROS 2, Linux, and embedded systems.

Education

University of Minnesota Twin Cities — *Minneapolis, MN, USA*
Master of Science (M.S.), Robotics — *Sept '24 – July '26*

Birla Institute of Technology and Science, Pilani (BITS-Pilani) — *Goa, India*
Bachelor of Engineering (B.E.), Electronics and Communication — *Aug '18 – May '22*

Technical Skills

Programming	C, C++ (Modern C++), Python, MATLAB, Bash
Tools & Frameworks	Git, Linux, Docker, CMake, Bazel, **Robotics** (ROS, ROS2, MoveIt!, PX4, RViz, Gazebo, Isaac Sim)
Hardware	Raspberry Pi, NUC, Nvidia Jetsons, Arduino, ESP32, STM32, Realsense cameras, Optitrack MoCap
Domain Interests	Robotics Software, System Integration, Trajectory Optimization, Motion Planning, Optimal Control

Experience

Komatsu Mining Corp — *Milwaukee, WI*
Robotics Software Intern — *June '25 – Aug '25*
- Developed a startup motor/encoder calibration algorithm in C++ for robotic excavator to eliminate manual motor homing
- Designed a human-machine teleoperation interface in C++ to control a robotic truck remotely using a steering wheel / pedal

Aerospace Department University of Minnesota Twin Cities — *Minneapolis, MN*
Research Engineer — *Jan '25 – May '25*
- Built a robotics research lab for the department consisting of Optitrack MoCap, PX4 UAVs, UGVs serving ∼500 students
- Implemented a trajectory optimization algorithm in MATLAB for environment monitoring using UAVs for wildfire scenarios

MARMot Lab, National University of Singapore — *Singapore*
Research Engineer — *Jan '22 – Aug '24*
- Developed a novel control/planning algorithm for environment inspection using legged robots via gaze optimization **(link)**
- Built an active mapping evaluation pipeline using ROS/Gazebo, RTAB-Map SLAM, and Realsense cameras (D435/T265)
- Coded an inverse kinematics-based control algorithm for hexapod omnidirectional locomotion using Python/ROS **(video)**
- Co-authored a RL-based solution (100x faster) to Multi-Robot Task Allocation. Tested on real Crazyflies **(ICRA paper)**

Selected Projects

Mars Rover for University Rover Challenge (Controls Team Lead)
- Designed path planning and tracking (PID, Stanley, Pure Pursuit) controllers for the manipulation and locomotion systems
- Developed ROS/C++ -based drivers and software interfaces to integrate sensors and motor drivers with Nvidia Jetson

Optimization-based Motion Planning for 3D Environments (code)
- Built a convex optimization–based collision-free trajectory generation framework in MATLAB using YALMIP/MOSEK
- Modeled cuboidal and cylindrical obstacles as convex sets using superellipse-based formulations for tractable optimization

Cooperative UAV-UGV localization using Extended Kalman Filter (EKF) (code)
- Developed a nonlinear UAV-UGV localization system in MATLAB using an EKF and simulation-based validation
- Evaluated estimator consistency using Monte Carlo testing with NEES/NIS analysis on simulated and real data

Dynamic Visual SLAM for Crowded Indoor Environments
- Integrated ORB-SLAM3 pipeline with Nvidia SegFormer to detect and reject moving objects in crowded environments
- Evaluated in real university environments, achieving 0.077 m ATE and 0.073 m / 1.33° RPE, showing robust localization

Clothbot - Cloth Manipulation using Self Supervised Value Network (poster, web)
- Developed a self-supervised value network policy using spatial action maps for dynamic cloth unfolding on a dual UR5
- Achieved 95% coverage on rectangular cloths and 87.68% on unseen garments (T-shirts) with zero-shot sim-to-real transfer

Flying Ad-hoc Network Simulator for multi-UAV exploration (code)
- Developed a co-simulation platform integrating NS3 and Gazebo through ROS/C++ for testing multi-UAV swarm tasks
- Implemented UAV swarm motion planning (C++) and analyzed network metrics like PDR, hop-by-hop and end-to-end delay
- Simulated a wildfire rescue UAV swarm (PX4 SITL and ROS) for surveillance application. **(ACM paper)**

Publications

- Dai, W., **Bidwai, A.**, & Sartoretti, G. (2024). *Dynamic Coalition Formation and Routing for Multirobot Task Allocation via Reinforcement Learning.* Published at **IEEE ICRA 2024**. **(paper)**
- Gong, Y., Sun, G., Nair, A., **Bidwai, A.**, Cs, R., Grezmak, J., . . . Daltorio, K. A. (2023). *Legged robots for object manipulation: A review.* Published in **Frontiers in Mechanical Engineering**. **(paper)**
- Dhongdi, S., Tahiliani, M., Mehta, O., Dharmadhikari, M., Agrawal, V., & **Bidwai, A.** (2022). *FANS: flying ad-hoc network simulator.* Published at **2022 ACM LANC** (Latin America Networking Conference). **(paper)**